

VAISHNAVI P. NICHAL

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OBJECTIVE

AI/ML undergraduate skilled in Python, Machine Learning, NLP, and Computer Vision with hands-on experience in real-world AI projects. Seeking opportunities in AI/ML Engineering, Data Science, Computer Vision, or NLP.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning | Sanjay Ghodawat University Expected 2026
CGPA: 8.67 / 10 (Till 6th Semester)

Higher Secondary (Class XII) — CBSE | Kendriya Vidyalaya No. 2, Belgaum 2021-2022
Percentage: 80%

Secondary School (Class X) — CBSE | Kendriya Vidyalaya No. 2, Belgaum 2019-2020
Percentage: 86%

TECHNICAL SKILLS

Languages: Python, C, C++, SQL
AI / ML: Scikit-learn, TensorFlow, Keras, OpenCV
NLP & Speech: NLTK, TF-IDF, Speech & Text Preprocessing
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Power BI
Databases: MySQL, MongoDB
Web: HTML, CSS, JavaScript, Node.js
Tools: Git/GitHub, Jupyter Notebook, Google Colab, VS Code
Additional: Networking, OOPs, DBMS, Ethical Hacking Concepts, Linux

INTERNSHIPS & CERTIFICATIONS

ML Intern | SNIT Technology LLP Jan 2026 – May 2026

- Developed AI/ML projects using Python, OpenCV, and Speech Processing.
- Built systems for occupancy detection, oil price prediction, and text summarization.

AWS APAC Solutions Architect — Virtual Internship | Forage June 2025

- Built scalable architecture using Elastic Beanstalk; documented cloud cost optimization strategies.

Tata Group — Data Analytics Job Simulation | Forage July 2025

- Performed EDA with GenAI tools to identify delinquency risk factors; structured data for predictive modeling.

PROJECTS

Offline Voice Assistant (Speech Recognition)

Developed an offline multilingual voice assistant using OpenAI Whisper for speech recognition and pyttsx3 for text-to-speech. Implemented audio normalization and push-to-talk interaction to improve recognition accuracy on CPU-based systems.

Lane Detection System using OpenCV

Built a real-time lane detection system using image processing (Canny edges, Hough transform) to identify lane boundaries in video frames. Designed to mimic ADAS lane assistance functionality; optimized for road curvature and noise handling.

AGRIGO AI — Smart Agriculture System

A crop recommendation, fertilizer prediction, and disease detection system integrated with a smart chatbot using the Gemini API. Takes NPK, rainfall, and temperature values as input to provide suitable agricultural recommendations.

Text Summarization Tool

Implemented an NLP-based summarizer using POS tagging and frequency analysis. Achieved concise summaries with minimal semantic loss across 100+ long-form text samples.

Hotel Management System

Designed a hotel system using a Python frontend and MySQL backend to manage bookings, check-ins, billing, and availability. Streamlined real-time room tracking with automated alerts.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Leadership Roles

- School Captain — Kendriya Vidyalaya No. 2 (2020 – 2022)
- Event Coordinator — College Annual Day & Fests
- House Captain — Green House

Hackathon Participation

- Competed in Smart India Hackathon 2024 — Intra-University Level

Sports & Cultural Activities

- Represented college in cricket, volleyball, badminton, and kho-kho.
- Anchored and participated in college fashion shows and dance events.
- NSS Volunteer — coordinated health drives, rural outreach, and tree plantation.

SOFT SKILLS

Problem Solving • Communication • Leadership • Time Management • Research & Documentation

LANGUAGES

English • Hindi • Marathi